Dorian Turner

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Education

Imperial College London *Oct. 2024 – June 2028*
MEng Computing

- **Degree:** 1:1 score achieved in Year 1 and 2. Predicted First Class (equiv. 4.0 GPA).
- **Relevant Modules:** Machine Learning - 86.42%, Software Engineering - 84.16%, Operating Systems - 82.98%.
- **Achievements:**
  Elected as **year-wide well-being representative**, liaising between students and department
  Most Interesting Extension, **ARMV8 C Project Prize** - Imperial College London
  Encode ARC '26 **1st Place Overall** - best on-chain liquidity provider track, won $1000
  People's Choice Award in Amazon University Engagement Program

Experience

Software Intern *London, England*
Bloomberg *June 2026 - Sept. 2026*

- Improved the performance of a **latency-critical distributed C++ microservice** in Bloomberg's Core Products division, responsible for validating data entitlements in the core data flow of the Bloomberg Terminal.
- Delivered a **5x speedup in the end-to-end** in the median latency case, and a **10x speedup** in the max latency case for the concerned tail ESG requests.
- Analysed service bottlenecks through **profiling and load testing** to identify inefficiencies and improve throughput in a high-scale distributed environment.
- **Technologies:** C++, Perf, Grafana, Locust, Strace, Flamegraph, Valgrind.

Software Intern *Oslo, Norway*
Bridge Technologies *July 2025 – Aug. 2025*

- Developed a machine learning based tool to derive MOS image scores for customers; it ran **15% faster** than the previous tool; was **32%** more correlated with human scores.
- Researched and developed a glitch detection algorithm to detect macro-blocking artifacts in live broadcast, to monitor thousands of streams.
- **Deployed LLMs** via a Slack chatbot for documentation Q&A and an FAQ website powered by a **local RAG** backend on a Mac Mini cluster.
- **Stack:** Python, OpenCV, FFmpeg, C++, ChatGPT, Local LLM, and FAISS.

Projects

FortDB *Git*, July - Aug. 2025

- Developed a **thread-safe, NoSQL, append-only** database in C with another student.
- Designed **custom abstract syntax trees (ASTs)** for hierarchical data storage, enabling versioned path resolution, subdocument nesting, and time-travel queries with efficient traversal.
- Implemented **concurrency and memory primitives**: thread-safe hashmaps with versioned linked nodes.
- **Tools:** C POSIX standard library, C, Nix, Linux, and NixOS

Slice *Git*, Aug. 2026

- Developed a Linux x86-64 **percentile profiler** that allows a deeper analysis of which **call-stack paths** occur during different percentile executions of a chosen **C++ function**.
- Targets a chosen **demangled ELF symbol** with **uprobes**, then combines **eBPF sampling**, invocation IDs, per-thread stacks, and percentile, thread, and time-range filters.
- **Reveals the slow path flamegraph in the p95:p100 case** for the execution of a function in a program (e.g. a running service) rather than just the aggregate.
- **Tools**: Rust, C++, DWARF, Linux, CMake, Ninja.

Technologies

Languages: C/C++, Java, Kotlin, Python, Haskell, Scala, Nix, SQL and Bash

Tools & Frameworks: Git, Docker, FFmpeg, OpenCV, FASTApi, Firebase, Claude Code, MCP and AI